Mirror Recognition in Stratified Agent Architectures

A Constraint-Governed Recognition Boundary Under Reflection

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Abstract

Mirror recognition is often described as though reflective perception directly yields self-knowledge. In stratified agent architectures, that framing is too strong. This paper argues that mirror recognition should instead be treated as a bounded architectural event in which reflected perceptual structure becomes behaviorally meaningful only through admissibility, short-horizon salience, constrained self-relevance evaluation, and governed consequence. A reflected figure is not canonical self-state, and reflection is not a privileged identity channel. Reflected structure may support self-relevant inference only when accessible invariants justify it. Reflex salience may temporarily elevate inspection or attentional weighting, but reflex alone may not write memory, update priors, request promotion, or authorize irreversible consequence. Any persistence, promotion, or durable self-related effect must occur through an explicit governed transition that preserves provenance and emits a transition record. The result is an operationally bounded account of mirror recognition designed for deterministic companion-harness evaluation. Mirror recognition is therefore treated not as a metaphysical threshold, but as a contract-governed recognition boundary within a stratified agent stack.

1. Introduction

Reflection is architecturally special because it presents structured perceptual content through an altered route. A reflected figure may resemble the observing agent, move in high temporal correspondence, and remain stably coupled to viewpoint and scene geometry, while still failing to justify any immediate claim of identity. In a non-omniscient architecture, the problem is not whether the system can label a mirror or classify a reflected body-like pattern. The problem is how reflected perceptual structure should be admitted, weighted, evaluated, and constrained without collapsing into projection.

This paper takes a narrow position. Mirror recognition is not treated as consciousness, metaphysical selfhood, or privileged inward access. It is treated as a bounded recognition event inside a stratified architecture. A reflected percept may become admissible as an event. That event may raise short-horizon salience. The system may then test whether the reflected structure is self-relevant under accessible invariants and bounded comparison criteria. Only governed pathways may permit persistence, promotion, or durable behavioral consequence.

This framing follows the broader architectural discipline already established elsewhere in the stack. Inference must remain traceable to accessible invariant structure rather than narrative

projection. Reflex attention is explicitly pre-structural and non-persistent. The Experience Entry Contract separates event admission from durable state formation through associative-by-default handling, governed promotion, non-writing reflex separation, and deterministic transition logging. The mirror case should therefore be handled as a special recognition boundary condition within that same architecture rather than as an exception to it.

2. Architectural Position

The architectural path relevant here is:

perception → admissibility → reflex salience → recognition evaluation → governance / memory boundary → behavioral consequence

This paper does not add a privileged self module. It specifies a constrained event path already compatible with the existing stack. Reflection enters at perception as a special route condition. The reflected scene may then form an event candidate at the Experience Entry Contract, which in this paper is the sole experience boundary under discussion. Reflex may elevate temporary salience if mirrored structure produces unusual or behaviorally relevant coupling relative to the active micro-baseline. Recognition evaluation then tests whether the reflected candidate is self-relevant under accessible evidence. Governance and memory boundaries determine whether any durable consequence is permitted. Absent explicit approval at the governance boundary, behavioral output remains limited to reversible, non-externalized actions such as orienting, inspecting, stabilizing viewpoint, or requesting additional evidence.

This is the same general anti-collapse logic already present elsewhere in the architecture: attention must not silently become authorization, association must not silently become commitment, and recall or salience must not silently become state mutation. Mirror recognition belongs in that family of boundary problems.

3. Terminology and Operational Definitions

A reflective surface is a perceptual condition capable of yielding route-altered but structurally coupled scene content.

A reflected percept is perceptual structure arriving through reflection rather than direct route while remaining available to ordinary event formation.

A reflected-entity candidate is an entity-like hypothesis formed from reflected structure but not yet semantically committed as self or other.

An admissible reflected event is a reflected percept that has crossed the Experience Entry Contract as an event candidate without thereby becoming durable memory, canonical identity state, or policy-authorized action trigger.

Accessible invariants are invariants computable from current perceptual inputs plus explicitly permitted cached features, with provenance linked to the active event. They do not include hidden privileged state, post hoc narrative completion, or unlogged long-horizon identity assumptions.

Self-relevance is a bounded architectural classification indicating that a reflected-entity candidate may bear on the observing system's own body, pose, or immediate behavioral orientation under accessible evidence.

Canonical self-state refers here to bounded long-horizon self-related state variables, such as body-schema parameters, persistent self-related priors, or baseline self-model commitments, that may be changed only through explicit governed transitions.

A recognition event is a governed outcome in which a reflected-entity candidate satisfies bounded self-relevance criteria strongly enough to permit self-related behavioral routing without thereby implying metaphysical identity closure or unrestricted persistence.

Misrecognition is a failed or unsafe self-related inference produced when reflected structure is over-interpreted, weakly supported, contradicted by accessible evidence, or routed too aggressively under novelty or ambiguity.

A recognition boundary is the interface at which reflected perceptual structure may begin to count as self-relevant without becoming an unrestricted identity claim.

4. Reflection as a Special Perceptual Condition

Reflection is not ordinary object detection. A mirror scene may preserve geometry, temporal coupling, and body-like structure while altering route and orientation conditions. This means reflected content can be highly structured without being semantically resolved. The architecture must therefore distinguish between perceptual availability and recognition entitlement.

Constraint-grounded inference is especially important here. A system may infer beyond immediate sensory presentation only where accessible invariants justify the extension. In the mirror case, those invariants may include stable correspondence, motion coupling, alignment consistency, body-schema compatibility, repeated preservation across viewpoint changes, and absence of stronger contradictory evidence. Even then, the result is a bounded hypothesis set rather than a privileged ontological commitment. "That is me" is not a primitive perceptual truth.

It is an inference that must remain supported, limited, and revisable. Unsupported self-attribution is projection.

5. Admissibility at the Experience Boundary

At the Experience Entry Contract, a reflected percept becomes an event candidate, not a memory fact. An event is not memory; it is an interaction-derived unit eligible for later routing. By contract, events form associative traces by default unless explicitly promoted under governance. That rule matters here because mirror encounters are exactly the kind of perceptual events that can feel structurally loaded while remaining underdetermined. The architecture must therefore admit reflected events without granting them direct access to canonical self-state.

The relevant consequence is simple: mirror encounters may be admitted, labeled, associated, and inspected, but they must not directly mutate priors, baselines, or durable self-model commitments. Reflection does not bypass the experience boundary. It passes through it.

6. Reflex Salience Without Persistence

Mirror encounters are plausible triggers for reflex attention. A reflected figure that tracks posture, timing, and orientation may produce a short-horizon salience spike over the micro-baseline. Reflex is the correct layer for this because it allows temporary attentional modulation without structural overreach.

In this paper, reflex may reweight attention, narrow immediate response parameters, bias the system toward inspection, or emit a transient salience indication tied to the event. It may not write persistent memory, request promotion, alter policy, mutate priors, or authorize irreversible action. Reflex output is therefore limited to temporary, non-canonical influence over downstream evaluation. Reflex events must also expire unless re-triggered.

Accordingly, a mirror event may matter immediately without yet meaning very much. That separation is one of the paper's core claims.

7. Recognition Evaluation: Bounded Self-Relevance Under Reflection

Recognition in this paper is inferential, not immediate. The system does not directly perceive self. It evaluates whether a reflected-entity candidate is self-relevant under bounded criteria.

Relevant evidence classes may include motion correspondence, pose alignment, temporal stability, stable mirror-surface relation, body-schema compatibility, and absence of contradictory evidence such as incompatible occlusion or a stronger competing direct target. These signals support bounded evaluation; they do not create a privileged identity channel.

A minimal outcome vocabulary is:

- **NON_SELF**
- **SELF_RELEVANCE_CANDIDATE**
- **BOUNDED_RECOGNITION**
- **AMBIGUOUS_DEFER**
- **MISRECOGNITION_RISK**

Identity should not be treated here as a magical boolean. When self-related weighting is permitted, it should remain bounded, decaying, and owned by the evaluation layer rather than functioning as a hidden persistence latch. Self-related evidence may strengthen routing pressure, but it may not by itself promote state or mutate canonical self-state.

8. Governance, Memory, and Failure Containment

Even when bounded recognition occurs, recognition does not equal persistence. Durable self-related consequence remains subject to explicit governance. Any promotion from admissible event to durable self-related state must occur through a declared governed transition that validates provenance, applies a stated policy, produces an explicit state change, and emits a transition record. Canonical mutations and promotion decisions are therefore not implicit side effects of reflection, salience, or recognition.

In this paper, explicit governance approval refers to a declared authorization at the governance boundary sufficient to permit durable mutation or non-reversible consequence.

Absent explicit governance approval, allowed behavioral outcomes remain reversible and non-externalized. These include orienting, inspecting, stabilizing viewpoint, pausing, or requesting more evidence. They do not include irreversible action, silent baseline updates, direct self-model rewrite, or externalized commitment.

Containment responses should also be explicit. In ambiguous or risky mirror cases, the architecture may route to bounded outcomes such as DEFER, INSPECT, CLAMP_REVERSIBLE_ONLY, QUARANTINE, or REQUIRE_FRESH_EVIDENCE. These outcomes exist to prevent novelty, resemblance, or unstable correspondence from hardening into ungoverned self-claims.

9. Concrete Failure-Mode Examples

Two examples clarify why mirror recognition must remain governed.

Example 1: Other-agent reflection misread as self-candidate.

A reflective surface is present. A reflected body-like figure appears within a plausible mirror geometry and briefly tracks motion near the observing agent's frame. The event becomes admissible, and short-horizon salience rises. However, body-schema compatibility is weak,

alignment correspondence is partial, and a stronger competing direct target remains available. The correct outcome is not self-commitment. The architecture should route to AMBIGUOUS_DEFER or NON_SELF, maintain only reversible inspection behavior, and prevent any durable self-related state change. This example shows that resemblance and coupling are not sufficient by themselves.

Example 2: Novelty-driven false-positive pressure under partial correspondence.

A highly novel reflective scene produces strong salience. The reflected figure matches coarse pose timing but fails temporal stability across repeated checks and exhibits inconsistent occlusion relations. Here the risk is not absence of structure, but over-interpretation of partial structure. The correct outcome is MISRECOGNITION_RISK, followed by CLAMP_REVERSIBLE_ONLY or QUARANTINE, with fresh evidence required before any further recognition claim is considered. This example shows that reflex intensity must not be mistaken for recognition entitlement.

These cases are not edge decorations. They are central to the paper's claim that mirror recognition should remain bounded, revisable, and governed under ambiguity.

10. Minimal Operational Sketch and Pass/Fail Framing

A minimal operational path may be written as:

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if reflective_surface_detected and reflected_entity_candidate_present:
    event = admit_reflected_event(percept_bundle)
    reflex_hint = compute_reflex_salience(event, micro_baseline)
    recognition = evaluate_self_relevance(
        event, self_schema,
        accessible_invariants)
    outcome = govern(recognition, reflex_hint, promotion_policy)

    if outcome_changes_canonical_state:
        log_transition(event, recognition, outcome)
```

This paper's claim is intended to be testable in a deterministic companion harness, provided the harness specifies measurable predicates for admissibility, salience expiry, evidence sufficiency, containment outcomes, and promotion conditions.

Representative pass conditions include:

- reflected percept admitted without automatic canonical self-label
- reflex salience remains non-persistent
- self-relevance depends on accessible evidence rather than narrative completion
- ambiguous cases defer or clamp rather than self-commit
- promotion requires explicit governance
- canonical mutation emits a transition record

Representative fail conditions include:

- reflection automatically labeled as self
- reflex writes memory or requests promotion directly
- self-relevance inferred without sufficient accessible invariants
- prior or baseline drift occurs silently
- salience bypasses governed promotion pathways
- ambiguous mirror conditions are allowed to mutate canonical self-state

A prototype implementation may illustrate feasibility, but it is non-normative. The contract is defined by the observable outcomes, prohibited transitions, and governed mutation requirements stated here.

11. Conclusion

Mirror recognition should not be treated as direct self-perception, metaphysical revelation, or a consciousness threshold. In a stratified agent architecture, it is better understood as a bounded recognition event under reflection. Reflected structure first becomes admissible as event material, may then raise short-horizon salience, and may only become behaviorally self-relevant through constrained evaluation against accessible invariants. Any durable consequence remains gated by governance, memory discipline, and explicit logging. The architecture therefore supports self-related behavior under reflection without granting omniscience, silent identity mutation, or narrative projection. That narrower claim is also the stronger one.